**Control Major Bleeding**

**General Care:**
- Expose and Support Injured Limb
- Check PMS

**Pain Management**

Go to Pain CPG

Dress Open Wounds as needed

Apply Appropriate Splinting Device

## Special Considerations

**Femur Fracture**
- Midshaft Femur Fracture – Apply Traction Splint
- Consider Traction Splint for distal third femur fractures
- Fractured femurs may result in significant blood loss, administer IV fluids as per traumatic shock CPG

EMT

**Anterior Shoulder Dislocation**
- Anterior shoulder dislocations without suspected fracture of the humorous, clavicle or shoulder joint may be reduced.
- Reduction should be achieved through gentle external rotation of the humorous. See discussion section.

AP

**Patella Dislocation**
- Medial or lateral patella dislocations without suspected fracture may be reduced.
- See discussion section

AP

**Patella Fracture**
- Patella fractures should be splinted in the position found

## Patient Care Goals

- Control external hemorrhage and prevent hemorrhagic shock
- Stabilize fractures and dislocations to minimize further injury and pain
- Manage pain effectively while maintaining patient safety
- Assess and maintain neurovascular function in affected limbs
- Facilitate timely transport to appropriate facilities for definitive care

## Patient Inclusion & Exclusion Criteria

**Inclusion Criteria**

- Patients with suspected or confirmed fractures, dislocations, or soft tissue injuries to extremities
- Traumatic amputations or open wounds with potential for significant bleeding
- Limb injuries from crush mechanisms or high-energy trauma
- Neurovascular compromise or compartment syndrome suspicion
- Scene information indicating mechanism likely to cause limb injury

**Exclusion Criteria**

- Limbs which are entrapped. (see Crush Injury CPG).

## Key Patient Assessment Points

1. **Initial Impression**
   - Scene safety, mechanism of injury, energy involved
   - Number of patients, need for additional resources
2. **Airway & Breathing**
   - Ensure airway patency, assess for associated chest trauma
   - Administer oxygen if $SpO_2$ <94% or respiratory distress
3. **Circulation & Perfusion**
   - Check for external bleeding, signs of shock
   - Evaluate pulses, capillary refill, skin color/temperature in affected limb
4. **Neurovascular Status**
   - Assess sensation, motor function, distal pulses
   - Check for deformity, swelling, crepitus, open wounds
5. **Compartment Syndrome Signs**
   - Pain on passive stretch, pallor, pulselessness, paresthesia, paralysis
6. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.
     - Red Priority: Life-threatening hemorrhage, hypotensive shock, neurovascular compromise.

## Patient Management & Interventions

1. **Primary Survey**
   - **Life-Threatening Injury Identification**
     - Detect immediate threats **using the C-ABC algorithm** (Catastrophic Hemorrhage, Airway, Breathing, Circulation).
     - Focus on catastrophic bleeding, airway obstruction, tension pneumothorax, or signs of shock (e.g., pallor, delayed capillary refill).
2. **Hemorrhage Control**
   - Apply direct pressure, **tourniquet if uncontrolled bleeding**
   - Use hemostatic agents for junctional wounds
3. **Secondary Survey**
   - Perform secondary survey as per Universal Care CPG.
   - Expose effected limb, assess for CMS:
     - Color
     - Movement

        - Sensation
4. **Pain Management**
    - Assess and administer analgesia per Pain Management CPG.
5. **Consider ALS back-up**
    - Consider the requirement for ALS back-up as per Universal Care CPG.
    - ALS back-up should be routinely requested for:
        - Severe pain management.
6. **IV Access & Fluid Resuscitation**
    - Manage shock as per General Trauma Management CPG.
7. **Splinting and Immobilization**
    - Splint fractures/dislocations in anatomic position.
    - Realign if neurovascular compromise, with analgesia.
8. **Limb Amputations**
    - Stump Care:
        - Cover the stump with a moist, sterile dressing.
        - Apply uniform, moderate pressure using an elastic or compression dressing to control bleeding.
        - Apply tourniquet if bleeding is not controlled.
    - Amputated Part:
        - Retrieve amputated part unless it will significantly delay transport.
        - Amputated parts should be wrapped in moist sterile dressings and placed in a plastic bag.
        - This should then be surrounded by a bag or container holding a mixture of ice and water.
        - Do not use ice alone and never use dry ice.
        - The amputated part must not be allowed to come into direct contact with ice for any length of time.
        - Cooling part slows chemical processes and will increase viability from 4 hours to up to 18 hours.
9. **Impaled Objects in Extremities**
    - Do not remove impaled objects in extremities as it could cause severe hemorrhage.
    - Unnecessary movement of the impaled object outside the body is translated or magnified with the tissues.
    - Secure the object in place with a bulky dressing.
10. **Femur Fractures**
    - Traction splints should be routinely applied to isolated mid-shaft femur fractures.
    - Consider ALS back-up for pain associated with applying the traction splint.
    - Application of a traction splint limits blood loss, it is appropriate to delay transfer to apply the traction splint, providing there are no other signs or symptoms that require immediate transfer.
    - Traction splints may be used on distal third femur fractures.
        - Due to the proximity to the knee joint, careful assessment is required prior to ensure no knee joint involvement.
11. **Dislocations**
    - Anterior shoulder dislocations and patella dislocations may be reduced, unless a fracture of the joint is suspected.
    - Other types of dislocations should be splinted in place.
    - Pain management for reduction: Penthrox is generally required as a minimum.
        - **Anterior Shoulder Reduction Technique**
            - Position the patient: Supine on a stretcher or sitting upright.
            - Step 1: Elbow Flexion and Traction
                - Flex the patient's elbow to 90 degrees.
                - Apply gentle, steady longitudinal traction along the arm, pulling distally (toward the feet in a supine patient) to disengage the humeral head from the glenoid rim.
                - Maintain traction throughout Step 2 to prevent muscle spasm.
            - Step 2: External Rotation
                - Slowly and gently externally rotate the forearm (move the hand outward, toward the patient's side) to about 70–90 degrees or until resistance is felt.

        - Pause if the patient experiences significant pain; allow muscles to relax.
        - Step 2 should be conducted slowly, allowing muscles to relax
    - **Lateral Patella Reduction Technique**
        - Place the patient supine on a stretcher with the hip flexed slightly (e.g., 20–30 degrees) to relax the quadriceps muscle.
        - Alternatively, the patient may sit with the leg supported if more comfortable.
        - Step 1: Knee Extension
            - Gently extend the knee by slowly straightening the leg. This relaxes the quadriceps muscle and aligns the patella with the femoral groove.
            - Apply mild pressure to the shin or foot to assist extension, avoiding forceful manipulation.
        - Step 2: Apply Medial Pressure
            - While extending the knee, place your hand or thumb on the lateral edge of the patella.
            - Apply gentle, steady pressure to push the patella medially (toward the midline) back into the femoral groove.
            - The patella often reduces with a palpable or audible clunk as it slides back into place.

12. **Patient Reassessment**
    - Frequently Reassess the patient as per Universal Care CPG.
    - Specific Patient Reassessment Points:
        - Limb CMS (colour, movement and sensation) must be reassessed:
            - Immediately after splinting.
            - Every 15 minutes.

## Patient Safety Considerations

- Ensure scene safety before approaching patient
- Avoid manipulation without analgesia if possible
- Reassess neurovascular status after splinting
- Use caution with tourniquets; document time applied
- Prevent hypothermia by covering patient
- Be prepared for deterioration during transport

## Documentation

- Identification: Age, gender, weight
- Medical complaint: Limb injury description, mechanism
- Injuries/Information: Affected limb, deformity, neurovascular status
- Signs: Vitals, pain score, bleeding assessment
- Treatment: Hemorrhage control, splinting, analgesia
- Allergies: If known
- Medications: Analgesics, fluids administered
- Background: Comorbidities, anticoagulants
- Other: Time of injury, amputated part management

## Clinical Insights & Additional Guidance

- The use of a tourniquet for uncontrolled limb bleeding is exceptionally important. Unsure that torniquets are applied when indicated.
- Manipulation of fractured limbs to apply a splint will likely increase pain. Ensure adequate analgesia is administered prior to splinting.

## Key Performance Indicators

1. Apply a traction splint for mid-shaft fracture femurs, when no time critical injuries are present.

## References

1. National Association of State EMS Officials (NASEMSO) (2022) National Model EMS Clinical Guidelines.
2. Queensland Ambulance Service (QAS) Clinical Practice Guidelines: Trauma/Limb injury.
3. Queensland Ambulance Service (QAS) Clinical Practice Guidelines: Trauma/Extremity injury.
4. St John New Zealand Clinical Procedures and Guidelines - Comprehensive edition.
5. BC Emergency Health Services (BCEHS) Clinical Practice Guidelines: Extremity Trauma.
6. Health Professions Council of South Africa Clinical Practice Guidelines Protocols.